

Interface Capability Brief

EDI 834 CSV Excel Conversion · Stakeholder overview · 2026-08-28

1. Executive summary

Enrollment teams often have CSV, tab-delimited TXT, or Excel — not X12. This demo picks up those files, maps them to HIPAA 834 enrollment XML, and writes a flat 834. The same mapping runs in reverse so an 834 becomes CSV for analytics.

2. Who this is for

- Sources: enrollment CSV / TXT (Excel is normalized to CSV in the Web UI on 23R1) - Destinations: X12 834 file; reverse path writes CSV - Demo vs production: synthetic members only. No SNIP gate, no real payer.

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: Directory / File listener
- Format: header row + member columns (MemberId, LastName, FirstName, BirthDate, GenderCode, RelationshipCode, MaintenanceTypeCode, PlanId, CoverageStartDate, sponsor/payer ids)
- Identity fields: MemberId (REF*0F)
- Samples path: samples/

Integration highlights

FQCNs are from PilotFish Documentation / PilotFish_V2 modules.conf. Runtime is V1 route.xml on pilotfish-eip:23R1.

Stage	Module / mechanism	Notes
Listener (fwd)	com.pilotfish.eip.modules.file.DirectoryListener	Polls input/inbound for .csv / .txt
Processors (fwd)	CSVTransformationProcessor → XSLTProcessor → EDITransformationProcessor	CSV/TXT → XML → 834 EDI XML → flat 834
Router (fwd)	XPathRoutingModule	Members present → 834; else kickout
Transports (fwd)	DirectoryTransport	Write .edi or kickout
Listener (rev)	DirectoryListener	Polls input/edi for .edi / .834
Processors (rev)	EDITransformationProcessor → XSLTProcessor → CSVTransformationProcessor	834 → XML → CSV XML → CSV
Transports (rev)	DirectoryTransport	Write .csv

4. Honest boundaries (not claimed yet)

- None for the first cut. SNIP-after-map can be added later like the tutorial mentions.

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - CSV TXT to EDI 834” from the Web UI or documented smoke path
- Exercise route “2 - EDI 834 to CSV” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- What is checked: at least one member row with MemberId or LastName - What is NOT checked: SNIP 1–7, full 834 situational rules - Does failure block outbound?: yes (kickout instead of 834)

6. State & idempotency

- Status model: file archive after poll (Move) - Dedup keys: inbound filename - Retry / poison: kickout folder; no automatic retry

8. Dual-write / side effects

- Order of commits: archive inbound, then write 834 or CSV - Compensation: none - Demo shortcuts: Excel .xlsx is converted to CSV in the Web UI. ExcelSheetProcessor is documented for 26R1.11 and is not assumed on the 23R1 image.

3. Inbound contract

- Transport: Directory / File listener - Format: header row + member columns (MemberId, LastName, FirstName, BirthDate, GenderCode, RelationshipCode, MaintenanceTypeCode, PlanId, CoverageStartDate, sponsor/payer ids) - Identity fields: MemberId (REF*0F) - Samples path: samples/

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - CSV TXT to EDI 834: runs steps such as Poll Enrollment CSV TXT, CSV to XML, Conditional Router, Map CSV XML to 834 EDI XML.

2 - EDI 834 to CSV: runs steps such as Poll EDI 834, EDI 834 to XML, Map 834 XML to CSV XML, XML to CSV.

7. Security & trust posture

- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Risk noted in design: Med — 23R1 vs 26R1 Excel module — Excel Sheet Converter deep-dive is 26R1.11
- Risk noted in design: Med — Trial X12 tables expired — Internal table data fails parse/render
- Risk noted in design: Low — Friendly names on large 834s — Tutorial mentions turning them off

8. Seeing it work

- - Ports: Web UI 8140, EIP 8139 - Compose project: edi-csv-excel-834 - Volumes: input/, output/, logs/, EDI/TableData/x12 → edi-tabledata - Heap: 512M–2048M (no SNIP) - Cold start: ~60–90s for EIP
- docker compose --profile full up -d --build
- - Local Web UI: http://127.0.0.1:8140/
- - EIP: http://127.0.0.1:8139/eip/

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.